

CSDA5110
Homework 4
Fall 1 2024

Reading Assignment for Week 4: Chapters 12, 13, 14 and 15

1. Load the midwest dataset from the ggplot2 package.
 - a. Use the filter function (page 161) to select observations in Ohio.
 - i. What is the average percentage of college educated among these counties?
 - ii. What is the average percentage of Asians in Ohio?
 - b. Use the group_by function to obtain average percentage of college educated and average percentage of Asians for all midwestern states.
2. Type the following code below into an Rscript file in order to create two data frames. Once you have created the data frames do the following:
 - Using commands found in the Tidyverse package, perform the following joins individually.
 - Left Join
 - Right Join
 - Full Join
 - Using comments in your script, explain the results of the output.

CODE TO CREATE DATA FRAMES

```
library(ggplot2)
data("economics")
```

```
econ2 <- economics %>% select(-date) %>% map_if(is.integer, ~.x/1000)
```

```
DF1 <- data.frame(wages1 = rlnorm(100, meanlog = log(50000)), pension1 =  
rlnorm(100, meanlog = log(1e+05)), hhid = 1:100)
```

```
DF2 <- data.frame(wages2 = rlnorm(100, meanlog = log(50000)), pension2 =  
rlnorm(100, meanlog = log(1e+05)), hhid = 1:100)
```